ML · #11

Predicting Metabolic Cost During Human-in-the-Loop Optimisation

Replacing a slow, lab-only face-mask measurement with wearable EMG and gait sensors, for exoskeletons that tune themselves outside the lab

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UE24CS352A Machine Learning · Mini-Project Review · October 2026

THE PROBLEM

Exoskeleton tuning is bottlenecked by a breathing mask

CMA-ES optimiser
proposes new torque profile

→

Exoskeleton
applies it for 2 min

→

Human walks
body adapts

→

Metabolic cost
measured via face mask

↻ *repeated 36× per 72-min session, feedback = metabolic cost*

~1 min
to settle after every change in control law, so it is slow feedback

≈ 5 %
irreducible noise in the measurement (MSE floor ≈ 0.0025)

Lab only
needs a face mask, a metabolic cart and fasting subjects

Our goal: learn f(EMG, gait, controls) → metabolic cost so the mask can eventually be replaced by sensors a person could wear anywhere.

DATASET

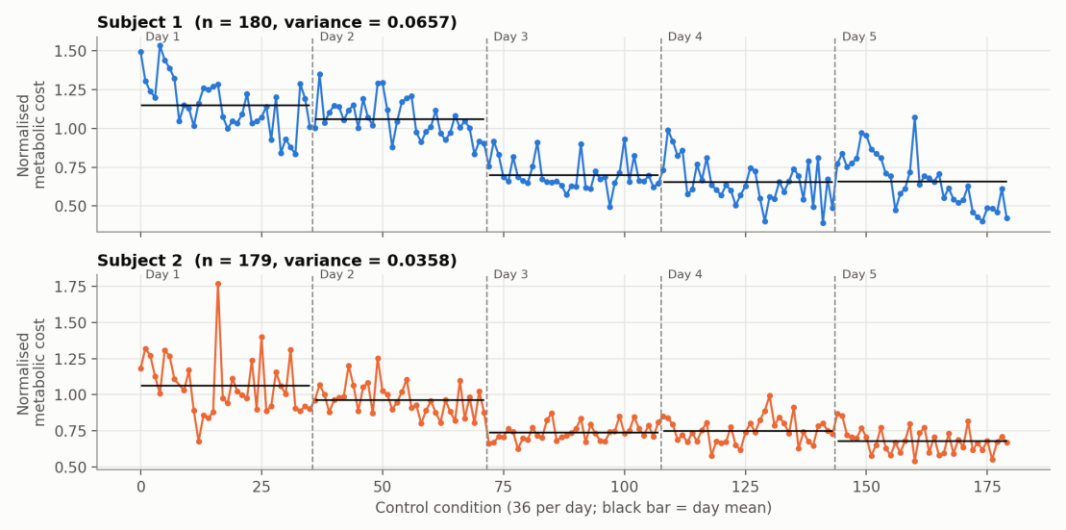
Real lab data: 2 people, 5 days, 36 assistance conditions per day

359
samples
(180 + 179)

29
features per
sample

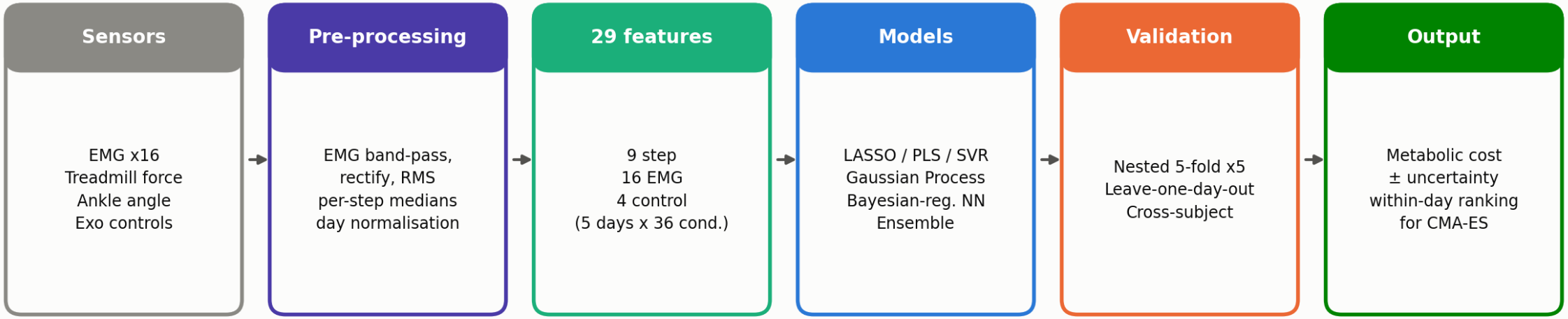
0.0025
noise-floor
MSE

- **16 EMG**
  8 muscles per leg; band-pass → rectify → RMS, normalised each day
- **9 Step / gait**
  peak ground force, ankle dorsi/plantar-flexion, step time, step width
- **4 Control**
  exo torque profile: peak torque, peak time, rise time, fall time
- **1 Target**
  metabolic rate - standing baseline, ÷ normal walking (1.0 = no exo)



METHODOLOGY

End-to-end pipeline



1 Leakage-free

Scaling and every hyper-parameter search happen inside the training folds (nested CV), so no test data influences model selection.

2 Two protocols

Repeated nested 5-fold CV for comparison, and leave-one-day-out to test generalisation to an unseen session.

3 Two layers

A faithful 1:1 Python port of the paper's MATLAB scripts (baseline), plus the extended study: 12 experiments, 16 figures, 10 unit tests.

We evaluate both predictive accuracy and whether it holds on a new day.

MODELS

13 models, including the paper's network rebuilt from scratch

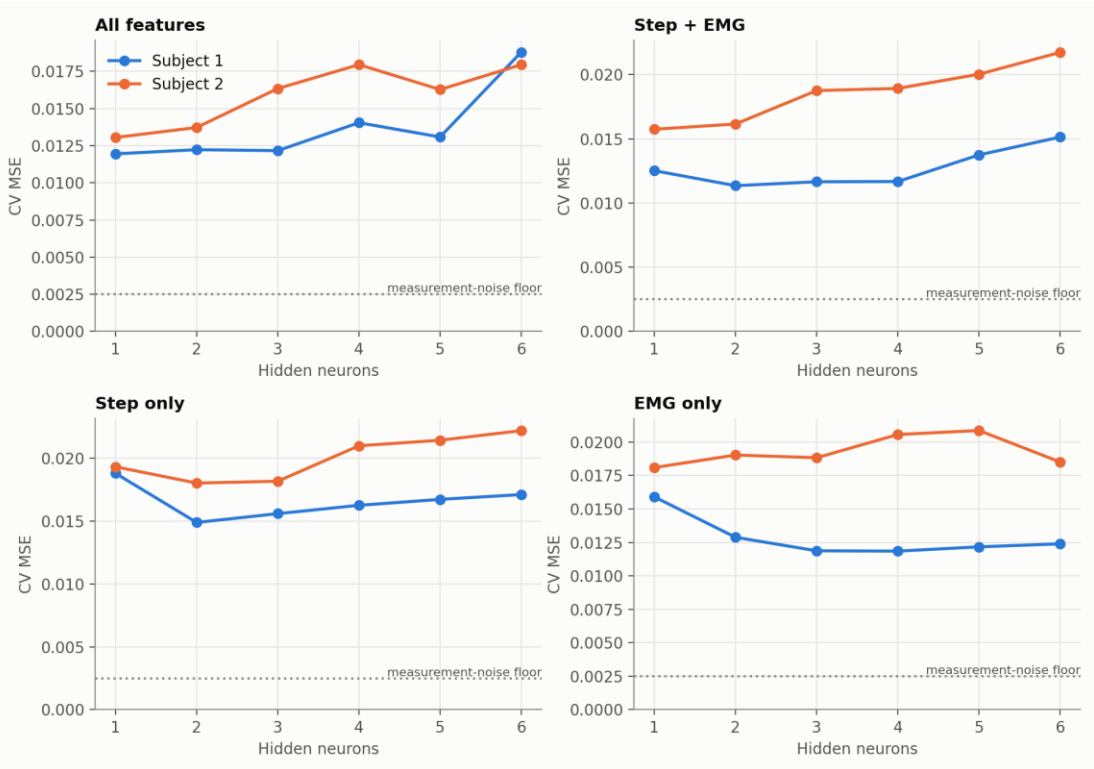
Linear	OLS · LASSO · Ridge · Elastic Net · PLS
Kernel / instance	SVR (RBF) · kNN · Gaussian Process
Trees	Random Forest · Extra Trees · Gradient Boosting
Neural	Bayesian-regularised NN (trainbr)
Ensemble	average of GP + SVR + LASSO + Extra Trees

Bayesian-regularised NN, from scratch

$F(w) = \beta \cdot E_D + \alpha \cdot E_W$

$\gamma = P - \alpha \cdot \mathrm{tr}(H^{-1}),\ \alpha = \gamma / 2E_W,\ \beta = (N - \gamma) / 2E_D$

Levenberg-Marquardt · closed-form Jacobian · picks 1-2 neurons



RESULTS · RANDOM CV

Our best models: lower MSE than the paper's references

0.0079

S1, ours (ensemble)
paper ref.: 0.0089 (lin. reg.)

R^2 0.88

S1 variance explained
(out-of-fold)

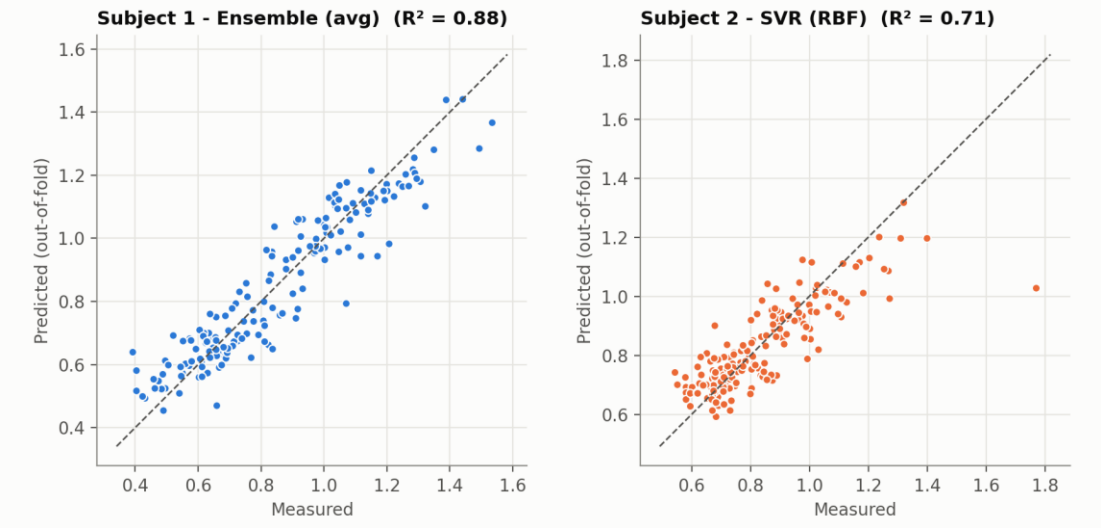
0.0109

S2, ours (SVR)
paper ref.: 0.0168 (lin. reg.)

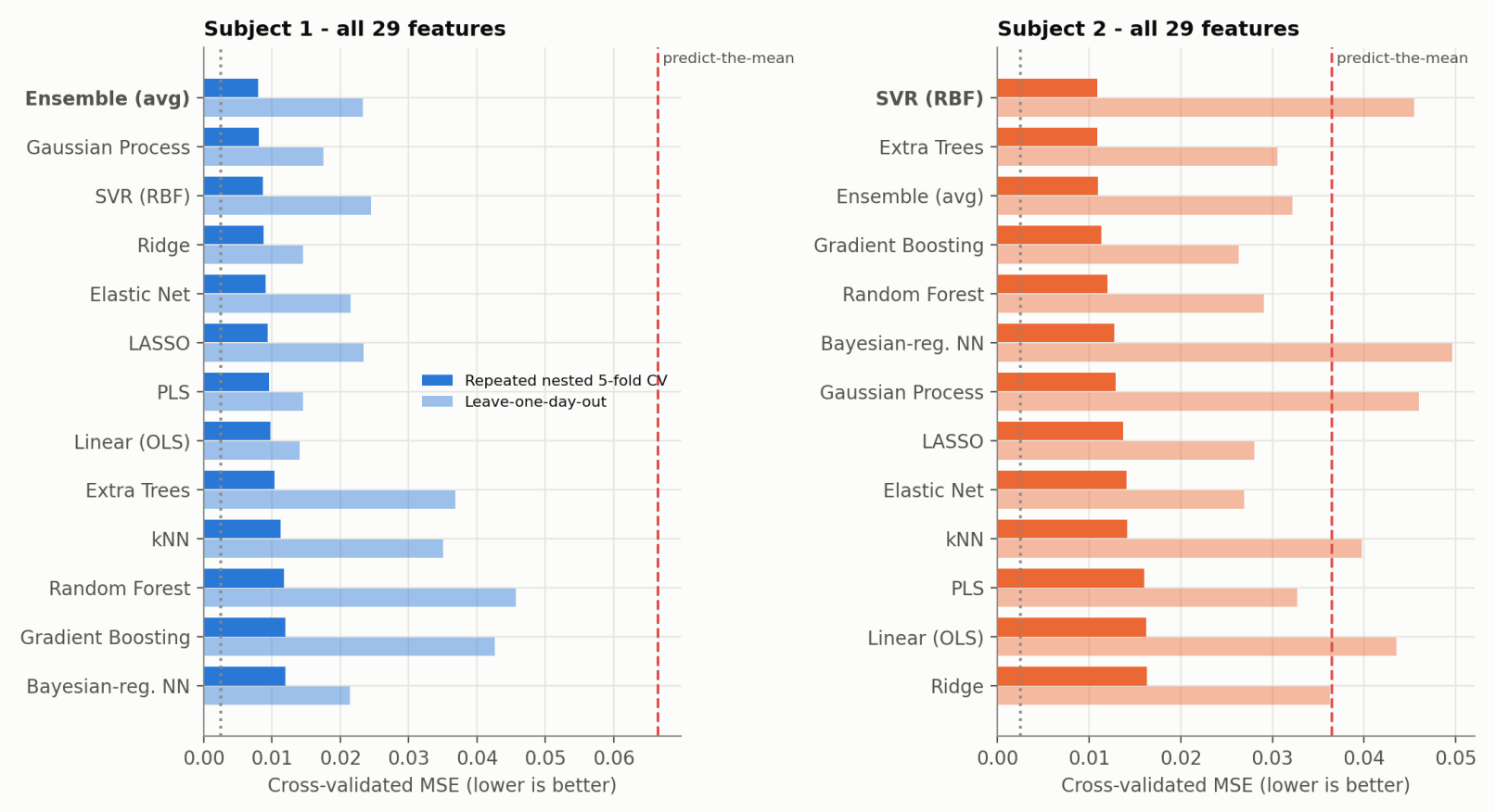
R^2 0.71

S2 variance explained
(out-of-fold)

Not a like-for-like comparison. Ours: best of 13 models, repeated nested 5-fold CV. Paper: linear regression under its own 10-fold CV. Unseen-day validation (next slides) shows the remaining challenge.



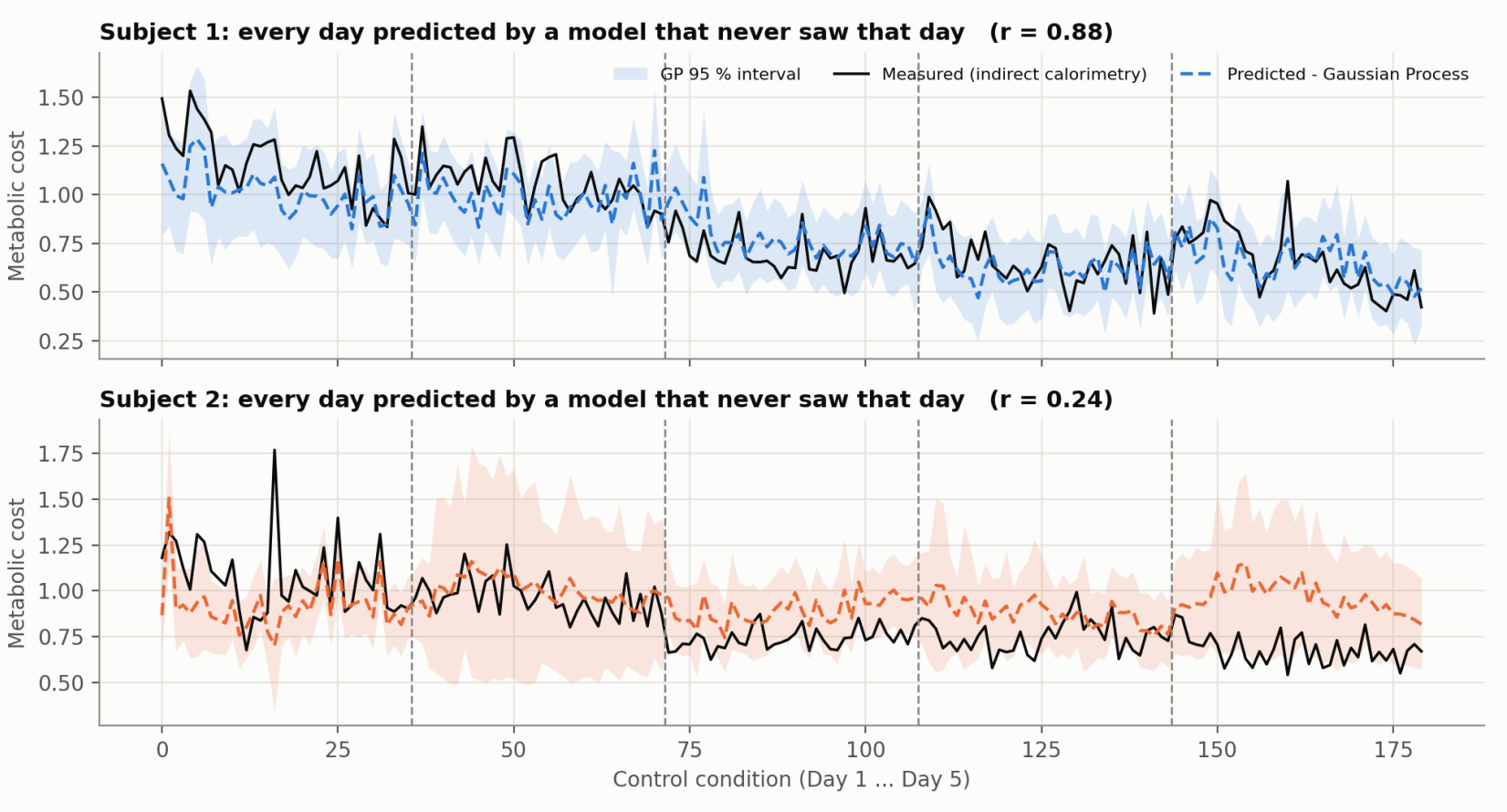
Random splits flatter models: unseen days are the real test



- **Random CV:** GP, SVR and the ensemble lead (MSE ≈ 0.008-0.011)
- **Unseen day:** error roughly doubles
- **Simple wins across days:** PLS / Ridge reach 0.0145 on S1
- **Trees & kNN** overfit within-day structure
- **Lesson:** report the protocol that matches deployment

RESULTS · GENERALISATION

Predicting a session the model has never seen



≈ 12 %

S1 unseen-day error, best model (OLS, MSE 0.014, R^2 0.79)

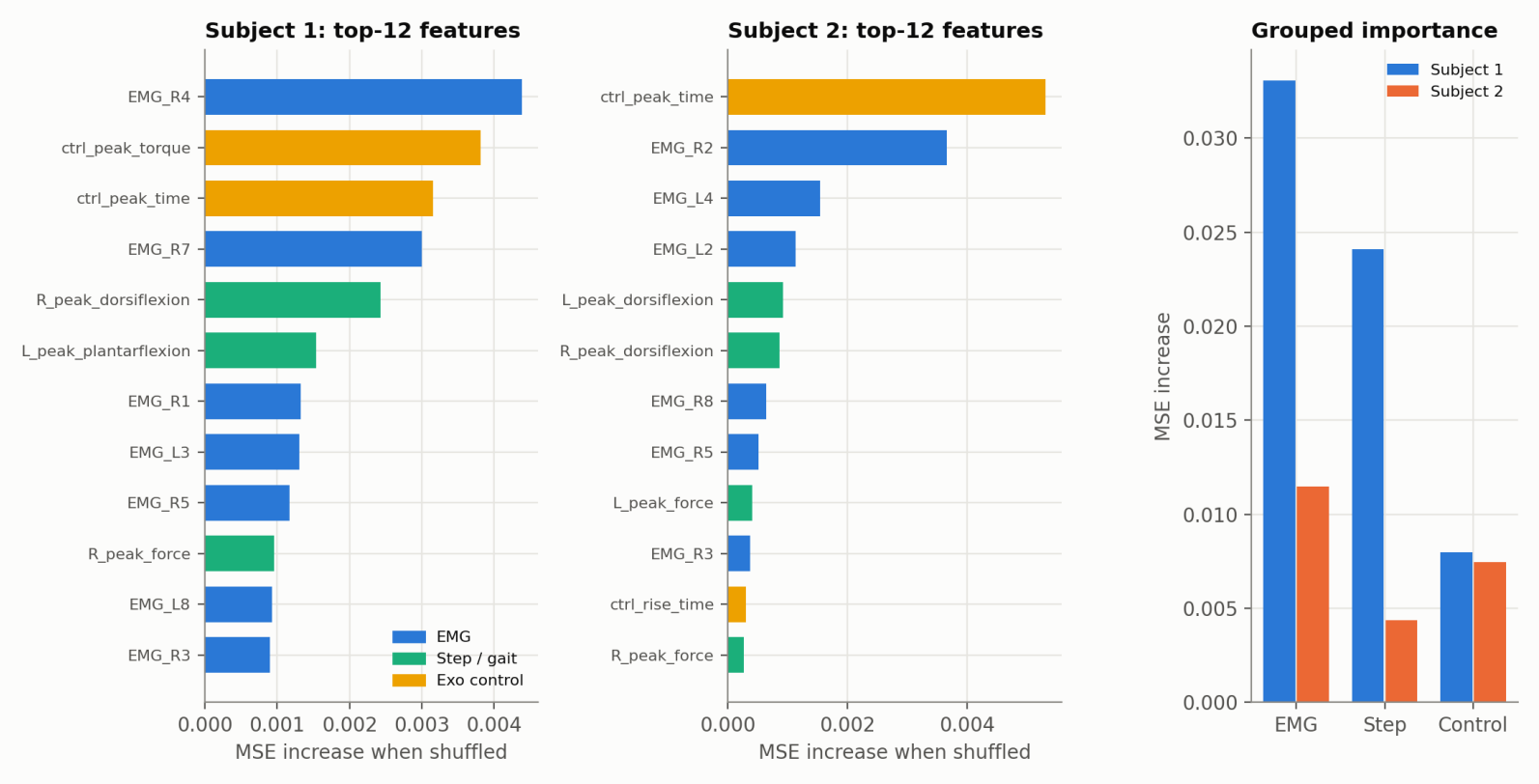
ρ ≈ 0.62

within-day rank correlation (S1), what the optimiser needs

≈ 16 %

S2 unseen-day error, best model (Grad. Boosting, MSE 0.026); GP shown: 0.046

EMG carries the signal, and a few muscles do most of the work



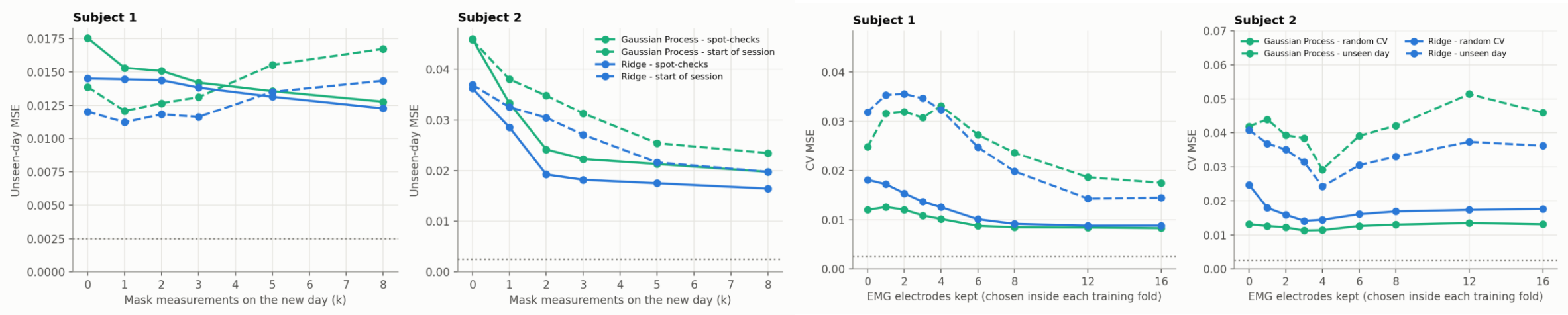
Feature set	S1 GP	S2 GP
All features	0.0081	0.0129
Step + EMG	0.0090	0.0120
EMG only	0.0107	0.0146
Step only	0.0151	0.0141

Feature-set ablation (CV MSE)

- **EMG is the strongest group in grouped importance** for both subjects. The ablation is a different test: for S2 (GP) step-only ≈ EMG-only (0.0141 vs 0.0146).

IMPROVEMENTS

Reducing unseen-day error with just 3 mask spot-checks



−51 %

S2 unseen-day error with just 3 of 36 mask spot-checks (GP 0.046 → 0.022)

$$w = k/(k+\sigma^2/\tau^2)$$

empirical-Bayes shrinkage: trusts calibration only as much as the training days justify

4 of 16

EMG electrodes beat all 16 for S2 (unseen-day Ridge 0.036 → 0.024)

DEMO

Live demo: the code, running

1 Baseline (paper port)

```
$ python
paper_reproduction/lin_reg_test.py


Linear Regression 5-Fold CV
Mean MSE: 0.010700
Mean R²  : 0.8349
```

2 Predict unseen days

```
$ python scripts/predict.py --demo
--subject S1


day      MSE   %err      r
  1   0.0286  16.9%   0.82
  2   0.0191  13.8%   0.49
  3   0.0132  11.5%   0.64
  4   0.0119  10.9%   0.61
  5   0.0149  12.2%   0.67
```

3 Automated tests

```
$ python -m pytest -q



.........
10 passed
```

1 = the paper's linear-regression baseline (S1 MSE 0.0107; paper reports 0.0089 with its own folds). Our best model (slide 6) reaches 0.0079. · Run from the ML proj folder, under a minute.

CONCLUSIONS & FUTURE WORK

Wearable signals can reduce dependence on metabolic measurements

9-10 % error within a session from EMG + gait (mean baseline: 19-26 %)

2× error on a new day; 3 mask spot-checks cut S2's by 51 % on this small dataset

EMG is the key signal; 4 of 16 electrodes beat all 16 for S2, so cheaper hardware is possible

1 : 1 models must be subject-specific; cross-person transfer is worse than the mean

Next steps

- Validate on more subjects, with anthropometrics + IMU
- Test spot-check calibration in real HILO sessions
- Use the surrogate as a prior inside CMA-ES / Bayesian optimisation

Thank you

Code, data, figures and the 2-page write-up are in the private GitHub repository.

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Ref: Krimsky & Ng, CS229 2018 · Zhang et al., Science 2017 · MacKay 1992 · Rasmussen & Williams 2006